

Module 5: Final Dataset Validation Audit

Executive Audit Summary

This document verifies that the feature engineering and data preprocessing pipeline executed successfully. The output dataset is verified to be complete, normalized, correctly encoded, and ready for K-Means / Hierarchical Clustering.

Validation Step	Target Standard	Observed Status	Audit Decision
Null Value Audit	0 Missing entries across all features	0 Nulls (Income imputed with median)	PASSED
Duplicate Check	0 Duplicate customer records	0 Duplicates detected in raw matrix	PASSED
Encoding Check	100% Numeric data types	Categoricals converted via One-Hot / Label	PASSED
Feature Scaling	Standardized (Mean=0, Std=1)	StandardScaler applied to all numericals	PASSED
Distribution Check	Minimized extreme skewness	Log1p applied on Income & Spending	PASSED
Dataset Export	Valid CSV export in data/processed/	final_engineered_customer_data.csv created	PASSED

Final Verification Metrics

- **Total Validated Records:** 2,240 rows
- **Total Features (Post-Encoding):** Transformed into fully numeric ML matrix
- **Storage Location:** `data/processed/final\_engineered\_customer\_data.csv`
- **Pipeline Status:** Certified for ML Model Training.